XRPL Validator — Daily Engineering Report · 2026-09-24

Period: 2026-09-24 00:00 to 24:00 US Eastern (2026-09-24 04:00 to 2026-09-25 04:00 UTC); ledgers #107,195,736 to #107,218,188. Every count from the ledger and the validator's logs stops at #107,218,188. Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: the day's first and last ledger, from a public full-history server; signing records; Step 1 receipt records (they begin 2026-08-29); deployment records (they begin 2026-09-11); xrpl-validator merge records (they begin 2026-07-04); amendment data (carried from the original report). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); Steps 2–5 receipt records (they begin 2026-10-09); data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Normal (agreement not checked) — coverage outside planned maintenance 100.000% (objective 99.9%), agreement with the network not checked (the reference node no longer holds these ledgers' hashes), no open engine receipt.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** signed 22,343 of the 22,453 ledgers the network closed (99.510%) (agreement not checked: the ledgers' hashes are no longer held). Unsigned: 110 during planned restarts.
- **Native engine:** 2 engine receipts recorded (2 fixed, 0 explained, 0 without a recorded fix).
- **Changes:** 1 deployment (cycle 154), 11 commits merged into main, 2 engine findings fixed (F399–F400).

Service levels

Measure	Value
Ledgers closed by the network	22,453
Validations issued	22,343
Coverage (validations / ledgers closed)	99.510%
Validations matching the network's validated ledger hash	not checked: the reference node no longer holds these ledgers' hashes
Unsigned during planned restarts	110
Unsigned outside maintenance	0
Coverage outside planned maintenance (objective 99.9%)	100.000%

Measure	Value
Restarts	1
Basis: the hash the first peer announced as accepted (before cycle 183)	22,343

Restarts:

- 22:58:52 UTC: deployment of cycle 154; validations resumed at #107,213,608 (110 ledgers unsigned between the last validation before the restart and the first after it)

The basis rows sum to validations issued. Coverage counts the validations the validator issued for ledgers the network closed in the period. Agreement compares each validation's ledger hash with the network's validated hash for that ledger. data.xrpl.org scores UTC days from the validations its crawlers received, so it can differ.

Native engine conformance

Ledgers with a receipt in the period (#107,195,736–#107,218,188), by step:

Step	Ledgers with a receipt
Step 1 · native engine	2
Step 2 · records	0
Step 3 · order	0
Step 4 · peer data	0
Step 4 · validations	0
Step 5 · own hash	0
Writer	0

Live counters are kept only for the current run; none exist for a past day.

Receipt register

ID	Time	Ledger(s)	Step	Description	Disposition
R-01	—	#107,210,679	Step 1 · native engine	1 transaction result(s) differed from rippled's; 11 change(s) of ours the network did not make; 1 object(s) differed in content	fixed: F399, commit ea76860, deployed in cycle 154 (2026-09-24 22:58 UTC)
R-02	—	#107,211,701	Step 1 · native engine	1 object(s) differed in content	fixed: F400, commit a08b8ee, deployed in cycle 154 (2026-09-24 22:58 UTC)

A receipt is recorded whenever the validator's own work differs from the network's, or the network's trusted validations reached it short of quorum. One ledger flagged by several steps is one receipt. Dispositions: fixed (with the finding, commit and deployment), explained (not a defect, with the reason), or open. Receipt and incident IDs are numbered per report: R-01 here is R-2026-09-24-01, INC-01 is INC-2026-09-24-01. Times are not shown where the reference node no longer holds the ledger.

Incidents

No incidents.

Change log

Deployments

Cycle	Main	Deployed	Test suites	Replay windows	Post-deploy health	Soak
154	`a08b8ee`	22:58:52 UTC	n/a	n/a	408 matched, 0 mismatched	#29 from 23:26 UTC: ended early: superseded by the next deployment (2026-09-25 14:14 UTC)

Cycle 154 delivered: `ef17d8f` fix(ledger): F398 — OfferStream original funds read through the afView's balance hooks (soak #27 receipt); `43fb741` docs: warm restart design — resume from the verified ledger bookmark, cold fallback; `88f4057` docs: warm restart design — incremental syncer off first, 1,000-ledger gap cap, deploy verdict timeout; `2d751ef` feat(node): warm restart — resume ticket, meta:last_seq bookmark and the resume checks; `2838f58` feat(node): warm restart — a stop gate that parks the validator's ledger feed at a ledger boundary; `c058ccf` feat(node): warm restart — the validator's ledger feed parks at a ledger boundary; every rollback rewinds meta:last_seq; `c8b4ad6` feat(node): warm restart — the resume sequence: ticket, bookmark, gap cap, root against the network; `a2518d1` feat(node): warm restart — clean stop writes a resume ticket; XRPL_WARM_RESUME starts from the verified bookmark; `99b7e0d` feat(node): resume_check — rehearse the warm-restart check on a kept state, read-only; `956ec00` fix(node): warm restart scans the wallet count before the validator's ledger feed adjusts it; `ea76860` fix(ledger): F399 — two-asset AMM withdrawals check the reserve before each new trust line (soak #28 receipt); `a08b8ee` fix(ledger): F400 — NFT IOU settlement grosses the transfer fee to nearest, as accountSend does (soak #28 receipt)

Each deployment passes the unit and port test suites and replays mainnet history windows with zero mismatches before it ships, is health-checked after it starts, and then soaks for 24 hours.

Merged into main

Merged	Repo	Commit	Subject
06:17 UTC	xrpl-validator	`43fb741`	docs: warm restart design — resume from the verified ledger bookmark, cold fallback
06:17 UTC	xrpl-validator	`88f4057`	docs: warm restart design — incremental syncer off first, 1,000-ledger gap cap, deploy verdict timeout
06:17 UTC	xrpl-validator	`956ec00`	fix(node): warm restart scans the wallet count before the validator's ledger feed adjusts it
06:17 UTC	xrpl-validator	`99b7e0d`	feat(node): resume_check — rehearse the warm-restart check on a kept state, read-only
06:17 UTC	xrpl-validator	`a2518d1`	feat(node): warm restart — clean stop writes a resume ticket; XRPL_WARM_RESUME starts from the verified bookmark
06:17 UTC	xrpl-validator	`c8b4ad6`	feat(node): warm restart — the resume sequence: ticket, bookmark, gap cap, root against the network
06:17 UTC	xrpl-validator	`c058ccf`	feat(node): warm restart — the validator's ledger feed parks at a ledger boundary; every rollback rewinds meta:last_seq
06:17 UTC	xrpl-validator	`2838f58`	feat(node): warm restart — a stop gate that parks the validator's ledger feed at a ledger boundary
06:17 UTC	xrpl-validator	`2d751ef`	feat(node): warm restart — resume ticket, meta:last_seq bookmark and the resume checks

Merged	Repo	Commit	Subject
20:12 UTC	xrpl-validator	`ea76860`	fix(ledger): F399 — two-asset AMM withdrawals check the reserve before each new trust line (soak #28 receipt)
21:15 UTC	xrpl-validator	`a08b8ee`	fix(ledger): F400 — NFT IOU settlement grosses the transfer fee to nearest, as accountSend does (soak #28 receipt)

Findings fixed

- **fix(ledger): F399 — two-asset AMM withdrawals check the reserve before each new trust line (soak #28 receipt)** (`ea76860`, xrpl-validator) rippled's `AMMWithdraw::withdraw()` runs `sufficientReserve` for each asset right before its send, in every withdrawal mode (AMMWithdraw.cpp 765/785): a withdrawal that would create a trust line for an account owning two or more objects needs base + increment x (OwnerCount + 1) against the balance before the fee. The single-asset paths had this check (finding 156); the two-asset equal withdrawals (tfWithdrawAll and …
- **fix(ledger): F400 — NFT IOU settlement grosses the transfer fee to nearest, as accountSend does (soak #28 receipt)** (`a08b8ee`, xrpl-validator) An NFTokenAcceptOffer settles in an IOU with rippled's `pay()` = `accountSend`, whose third-party send grosses the sender's debit with `multiply(saAmount, transferRate)` (TokenHelpers.cpp:869): STAmount multiply, `Number{v1} * Number{v2}`, rounded to the NEAREST 16-digit value. The engine used `gross_in`, the payment engine's round-up `mulRatio` (correct for BookStep, not for accountSend).

Other engineering changes

- **fix(node): warm restart scans the wallet count before the validator's ledger feed adjusts it** (`956ec00`, xrpl-validator) Both cold startup branches scan the AccountRoot count; the warm branch did not, so the counter started at 0 and the first deleted AccountRoot wrapped it to 2^64 - 1 on the dashboard. run_warm_resume now scans it beside the hasher rebuild (no added restart time). Found by the final whole-branch review.
- **feat(node): resume_check — rehearse the warm-restart check on a kept state, read-only** (`99b7e0d`, xrpl-validator)
- **feat(node): warm restart — clean stop writes a resume ticket; XRPL_WARM_RESUME starts from the verified bookmark** (`a2518d1`, xrpl-validator)
- **feat(node): warm restart — the resume sequence: ticket, bookmark, gap cap, root against the network** (`c8b4ad6`, xrpl-validator)
- **feat(node): warm restart — the validator's ledger feed parks at a ledger boundary; every rollback rewinds meta:last_seq** (`c058ccf`, xrpl-validator)
- **feat(node): warm restart — a stop gate that parks the validator's ledger feed at a ledger boundary** (`2838f58`, xrpl-validator)
- **feat(node): warm restart — resume ticket, meta:last_seq bookmark and the resume checks** (`2d751ef`, xrpl-validator)

Docs and operations

- **docs: warm restart design — resume from the verified ledger bookmark, cold fallback** (`43fb741`, xrpl-validator)

- **docs: warm restart design — incremental syncer off first, 1,000-ledger gap cap, deploy verdict timeout** (`88f4057`, xrpl-validator)

Independence program

No independence record for UTC day 2026-09-24; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

Carried over unchanged from the original report, which read it from the XRP Ledger when it was generated; the reference node no longer holds this day's ledgers.

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Voting events:

- PermissionDelegationV1_1 gained majority — ledger 107,212,033, 2026-09-24 21:25 UTC

Amendment	Majority since	Activation due
BatchV1_1	2026-09-15 14:06 UTC	2026-09-29 14:06 UTC
PermissionDelegationV1_1	2026-09-24 21:25 UTC	2026-10-08 21:25 UTC

Amendments enabled on mainnet: 94.

Open items

None.

Notes

Hand-written notes on the day's XRPL work: research and operations.

- **Warm restart (merged, not yet deployed).** Every deployment has restarted the node from an empty store and re-downloaded the full ledger state (about 19.9 million objects), so it resumed 90–111 ledgers behind the network. The node can now resume from its own saved state instead: a clean stop parks the ledger-sync loop between ledgers and records the last verified ledger; on restart the node checks that its stored state names that ledger, rebuilds the state tree and requires its root to equal the network's account hash for that ledger before continuing. Any doubt falls back to the full resynchronisation. Earlier attempts to restart without resynchronising (April and August) failed because they resumed from the wrong ledger; this design verifies the ledger first. A rehearsal on

two real mainnet states kept from the 2026-09-23 deployments rebuilt both to the network's account hash (31–34 s each). The first deployment that carries this code still starts cold; the one after it is the first warm restart.

- **Soak #28 receipt and finding F399.** At ledger 107,210,679 an AMMWithdraw (withdraw all, from a BTC/xSPECTAR pool) came from an account holding 2.799986 XRP at an owner count of 9 with no trust line for either pool asset. rippled checks the reserve for each new trust line before paying each asset out, in every withdrawal mode; a new line at that owner count needs 3 XRP, so mainnet returned `tecINSUFFICIENT_RESERVE`. The engine's two-asset withdrawal paths (withdraw all, and withdrawal by LP-token amount) skipped that check and paid both assets out. Fixed (F399) with the mainnet transaction pinned as a regression vector; verified and merged; deployment awaits a decision.
- **Operations.** At 01:40–01:55 EDT a test build on the machine that runs the reference rippled node pushed it into memory pressure; its validated ledger fell about ten minutes behind and the live node paused until it recovered (no mismatches, no engine receipts). Full test batteries no longer run on that machine while the live node depends on it.
- **Second soak #28 receipt and finding F400.** At ledger 107,211,701 an NFTokenAcceptOffer settled a buy offer priced 211,299.047746283 SOLO; the SOLO issuer charges a 0.01% transfer fee, so the buyer's debit is the price times 1.0001. rippled grosses that debit with its standard amount multiplication, rounded to the nearest 16-digit value (211,320.1776510576); the engine used the payment engine's round-up rule for the same product (…0577), leaving the buyer's SOLO balance one unit lower in the sixteenth digit. Earlier NFT sales with a transfer fee had exact products and could not show the difference. Fixed (F400) with the mainnet transaction pinned as a regression vector; verified and merged; deployment awaits a decision.
- **Deployment (F398–F400 and the warm restart).** Build a08b8ee replaced the live shadow build at 18:58 EDT after its gate (1,621 suite tests and 707 flow-engine tests passing) and both mainnet replay windows (60 and 98 ledgers, zero mismatches). This deployment still started from a full resynchronisation, as designed (the previous build could not leave a resume ticket). The health check matched 408 ledgers with no mismatches, and soak #29 began at 19:26 EDT (seed ledger 107,213,499). The receipt canary confirmed the comparison is live on the new build.
- **Build infrastructure.** Test gates now run on a dedicated build machine instead of the machine that hosts the reference rippled node, whose disk the test builds had been saturating. The gate completed in 10 minutes (previously about 15) with no effect on the reference node.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Vault-Scoped Delegated Session Keys** (custody) Add a SessionKey ledger object (ltSESSION_KEY, owner-reserved, keylet::sessionKey(account, sequence)) created via a new SessionKeySet transaction carr

- **Vault Collateral Clawback Settlement** (lending) Add an lsfVaultClawbackable flag to the Vault SLE, settable only at VaultCreate when the vault's underlying asset issuer holds clawback rights (AMMCla
- **Credential-Discounted Vault Collateral Ratios** (lending) Add sfCreditCredentialType and sfMaxUncollateralizedBps SFields to LoanBroker/Vault creation, letting a vault owner opt in to accepting a specific Cre

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
09-24 04:35	09-24 06:17	xrpl-validator	`43fb741`	docs: warm restart design — resume from the verified ledger bookmark, cold fallback
09-24 04:39	09-24 06:17	xrpl-validator	`88f4057`	docs: warm restart design — incremental syncer off first, 1,000-ledger gap cap, deploy verdict timeout
09-24 06:16	09-24 06:17	xrpl-validator	`956ec00`	fix(node): warm restart scans the wallet count before the validator's ledger feed adjusts it
09-24 06:16	09-24 06:17	xrpl-validator	`99b7e0d`	feat(node): resume_check — rehearse the warm-restart check on a kept state, read-only
09-24 06:16	09-24 06:17	xrpl-validator	`a2518d1`	feat(node): warm restart — clean stop writes a resume ticket; XRPL_WARM_RESUME starts from the verified bookmark
09-24 06:16	09-24 06:17	xrpl-validator	`c8b4ad6`	feat(node): warm restart — the resume sequence: ticket, bookmark, gap cap, root against the network
09-24 06:16	09-24 06:17	xrpl-validator	`c058ccf`	feat(node): warm restart — the validator's ledger feed parks at a ledger boundary; every rollback rewinds meta:last_seq
09-24 06:16	09-24 06:17	xrpl-validator	`2838f58`	feat(node): warm restart — a stop gate that parks the validator's ledger feed at a ledger boundary
09-24 06:16	09-24 06:17	xrpl-validator	`2d751ef`	feat(node): warm restart — resume ticket, meta:last_seq bookmark and the resume checks
09-24 20:11	09-24 20:12	xrpl-validator	`ea76860`	fix(ledger): F399 — two-asset AMM withdrawals check the reserve before each new trust line (soak #28 receipt)
09-24 21:15	09-24 21:15	xrpl-validator	`a08b8ee`	fix(ledger): F400 — NFT IOU settlement grosses the transfer fee to nearest, as accountSend does (soak #28 receipt)

Times in UTC. A commit appears on the day it was committed to main or merged into it.